INTERIM FEDERAL SPECIFICATION

ENAMEL, SEMI-GLOSS, FOR METAL AND WOOD FURNITURE

This amendment was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. The General Services Administration has authorized the use of this amendment as a valid exception to Interim Federal Specification TT-E-001793, dated May 11, 1972.

PAGE 2

Paragraph 2.2, under "American Society for Testing and Materials (ASTM) Standards", add "D3335 - Determination of Low Concentration of Lead in Paints by Atomic Absorption Spectroscopy

PAGE 4

Table I, line 5, change to read: "Lead Content (Percent by Weight of Nonvolatile)." Under maximum, add 0.06 percent.

PAGE 8

Table II, last line, add in appropriate columns: Lead content - D3335 - 4.2.2.5.

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Insert the following new paragraphs:

"4.2.25 Lead content. The method described in 4.2.25.1 or the method described in 4.2.25.2, shall be used. The X-ray fluorescence spectrometry method described in 4.2.25.1 shall be the final determinant of compliance in all cases.

4.2.25.1 Analysis for lead by X-ray fluorescence spectrometry. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight. The molybdenum X-ray source shall be operated at 60Kv and 54Ma; a lithium fluoride crystal cut along the 200 planes shall be used to disperse fluorescent X-rays; the detector collimeter shall be set at fine; and flow and scintillation counters shall be used as detectors. Pulse height selection shall be used in all measurements, and the counting time shall be 100 seconds. Place the sample disc in the sample holder, and measure the count rates of lead, lead background, and the molybdenum Rayleigh-scattered background from the X-ray tube at the following angles:

Analytical Line	Angle	Intensity
Mo	20.33	I_{Mo}
Background I	33.50	I'
Pb	33.93	I_{Pb}
Background II	34.50	I"

4.2.25.1.1 Calculation. Determine the ratio R of the lead and molybdenum lines as follows:

$$R = \frac{2I_{Pb} - I' - I"}{2I_{Mo}}$$

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4.25.1.2 Procedure. Duplicate drawdowns of the well-mixed paint shall be made on the sealed portion of a standard paint penetration chart, using a mechanical applicator plate and a film applicator with a 150 m gap. The drawdowns shall be at least 250 mm long, and shall be allowed to air-dry for 24 hours at standard conditions. Discs shall then be cut from each drawdown to fit the sample holder of an X-ray fluorescence spectrometer. Standard samples of the same type of paint containing known amounts of lead shall be prepared and measured in the same way, and the R values shall be plotted against the lead concentration to obtain a working curve. By use of the working curve, the value of R obtained for the test sample shall be converted to the lead concentration in percent by weight, and the result shall be evaluated for compliance with the requirement in table I.

4.2.25.2 Analysis for lead by Atomic Absorption Spectrometry. ASTM Method D3335 shall be used to determine the concentration of lead in percent by weight, and the result shall be evaluated for compliance with the requirement in table II."

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